

Onboard Data from Syslog-ng Server

Summary

This document includes the steps involved in Onboarding data from Syslog-ng to Intermediate Forwarder by Creating Data Inputs in Intermediate Forwarder and ensuring the required port is open in Firewall, Configuring data flow on syslog-ng to Intermediate Forwarder via the enabled port and Verifying on Splunk Enterprise.

1. Install syslog-ng

Refer **02_Install_and_configure_syslog-ng_Linux_V1.pdf**.

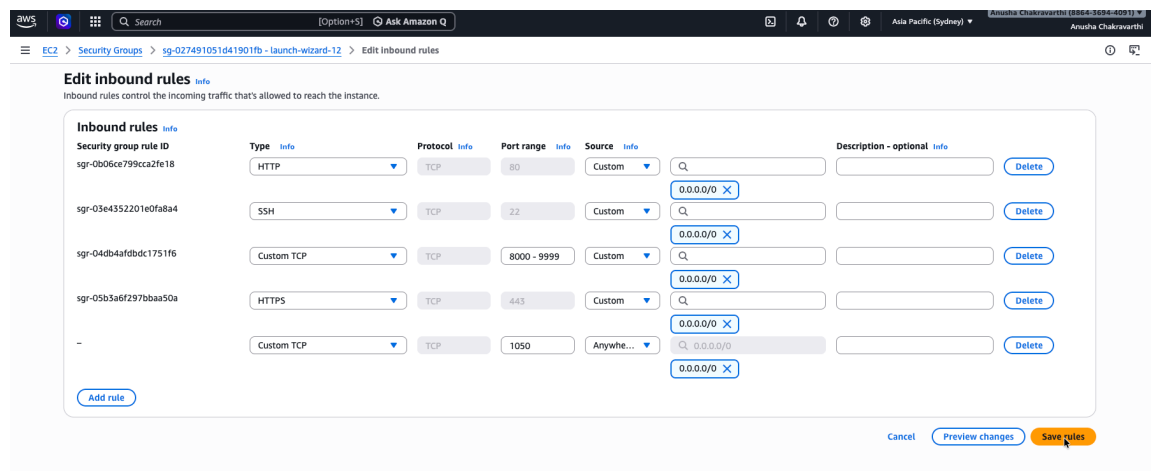
2. Create data inputs for TCP ports in Intermediate Heavy Forwarder

1) Ensure the port 1050 (TCP) is open on the Server Firewall (AWS Security Group)

Select the HeavyForwarder instance > Security tab > Security group (sg-xxxx) > Edit Inbound rules > Add rule

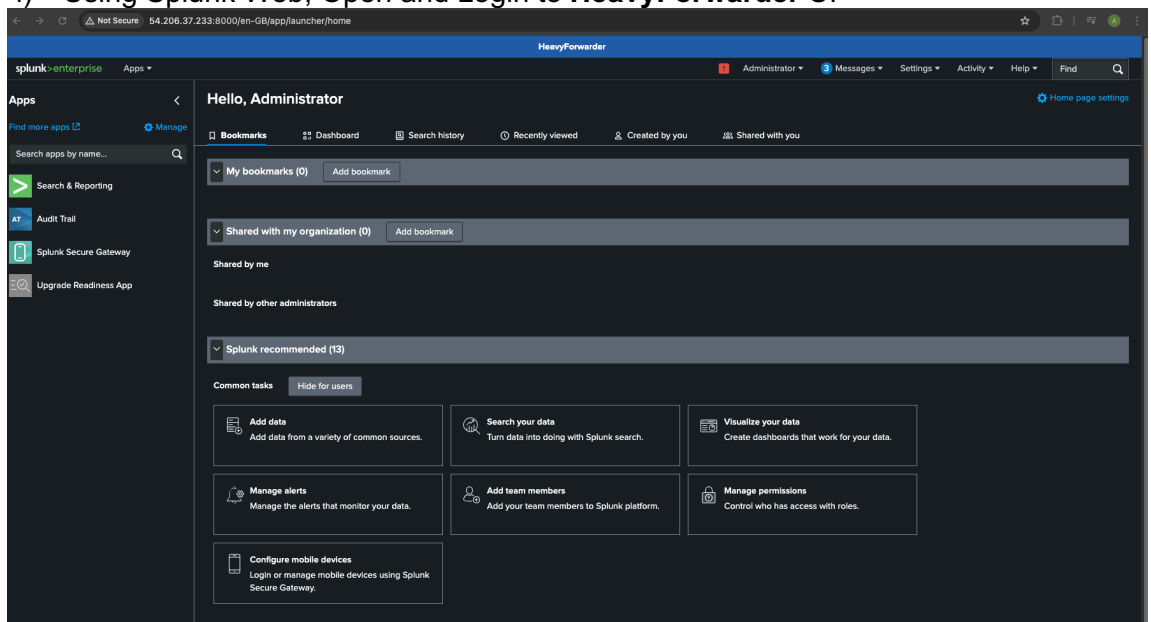
Type the following lines:

```
Type: Custom TCP
Protocol: TCP
Port range: 1050
Source: Custom, 0.0.0.0/0
```

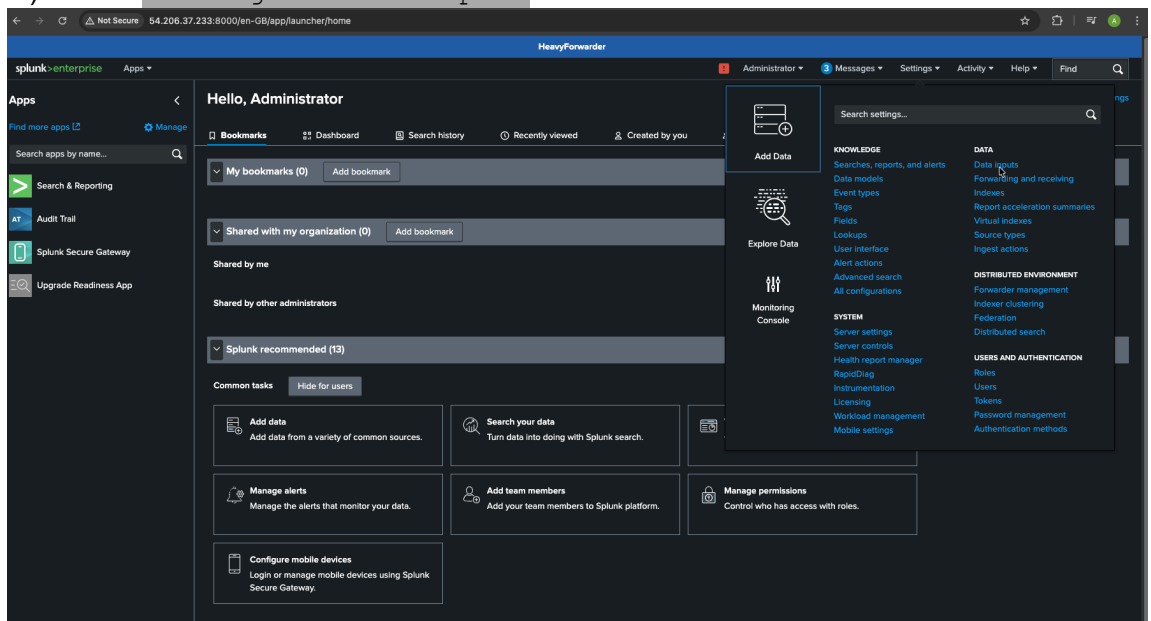


2) Configure Data Inputs in Intermediate (Heavy) Forwarder

i) Using Splunk Web, Open and Login to HeavyForwarder UI



ii) Go to Settings > Data Inputs



iii) Click TCP > Click New Local TCP

Data inputs

Set up data inputs from files and directories, network ports, and scripted inputs. If you want to set up forwarding and receiving between two Splunk instances, go to [Forwarding and receiving](#).

Local inputs

Type	Inputs	Actions
Files & Directories Index a local file or monitor an entire directory.	19	+ Add new
HTTP Event Collector Receive data over HTTP or HTTPS.	0	+ Add new
TCP Listen on a TCP port for incoming data, e.g. syslog.	0	+ Add new
UDP Listen on a UDP port for incoming data, e.g. syslog.	0	+ Add new
Scripts Run custom scripts to collect or generate more data.	37	+ Add new
Systemd Journal Input for Splunk This is the input that gets data from journald (systemd's logging component) into Splunk.	0	+ Add new
Logd Input for the Splunk platform This input collects data from logd on macOS and sends it to the Splunk platform.	0	+ Add new
Splunk Secure Gateway Initializes the Splunk Secure Gateway application to talk to mobile clients over websockets	1	+ Add new
Splunk Secure Gateway Mobile Alerts TTL Cleans up storage of old mobile alerts	1	+ Add new
Deep Link Dashboard Modular Input Initializes the Deep Link Dashboard Modular Input to complete registrations	1	+ Add new
Splunk Secure Gateway Deleting Expired Tokens Delete expired or invalid tokens created by Secure Gateway from Splunk	1	+ Add new
Splunk Secure Gateway Role Based Notification Manager Used for sending mobile alerts to users by role	1	+ Add new
Splunk Secure Gateway Enable Determine if Splunk Secure Gateway core modular inputs should be enabled	1	+ Add new
Splunk Secure Gateway KVStore Upgrade Updates kvstore collections to the proper state	1	+ Add new
Splunk Secure Gateway Metrics Collector Collects metrics for Splunk Secure Gateway	1	+ Add new

Showing 1 of 14 modular inputs

« Prev 1 2 Next »

iv) Enter port number: 1050

The screenshot shows the 'Add Data' configuration page for a Splunk HeavyForwarder. The 'TCP / UDP' option is selected. The 'Port' field is set to 1050. The 'Source name override' and 'Only accept connection from' fields are optional.

v) Under **Set Sourcetype**, choose **syslog** (or create a custom sourcetype, ex: syslog_tcp) Add optional host, source overrides if needed

The screenshot shows the 'Input Settings' page for a Splunk HeavyForwarder. The 'Source type' dropdown is open, showing 'syslog' as the selected option. The 'Index' dropdown is set to 'Default'.

vi) **Set Index:** Choose an index like `main` or a dedicated one like `syslog_idx`

Input Settings

Optionally set additional input parameters for this data input as follows:

Source type
The source type is one of the default fields that the Splunk platform assigns to all incoming data. It tells the Splunk platform what kind of data you've got, so that the Splunk platform can format the data intelligently during indexing. And it's a way to categorize your data, so that you can search it easily.

App context
Application contexts are folders within a Splunk platform instance that contain configurations for a specific use case or domain of data. App contexts improve manageability of input and source type definitions. The Splunk platform loads all app contexts based on precedence rules. [Learn More](#)

Host
When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Index
The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

FAQ
 > How do indexes work?
 > How do I know when to create or use multiple indexes?

vii) Click **Review > Submit**

Review

Input Type TCP Port
 Port Number 1050
 Source name override N/A
 Restrict to Host N/A
 Source Type syslog
 App Context launcher
 Host (IP address of the remote server)
 Index main

viii) Click **Start Searching**

✓ **TCP input has been created successfully.**
 Configure your inputs by going to [Settings > Data Inputs](#)

Start Searching Search your data now or see examples and tutorials. [🔗](#)

Extract Fields Create search-time field extractions. [Learn more about fields.](#) [🔗](#)

Add More Data Add more data inputs now or see examples and tutorials. [🔗](#)

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3. Configure Syslog-ng to send data via TCP port (1050) to Splunk

- 1) Login as root user in Syslog server

```
sudo su
```

- 2) Enable syslog

```
systemctl enable syslog-ng
```

```
[root@ip-172-31-0-46 ec2-user]# systemctl enable syslog-ng
[root@ip-172-31-0-46 ec2-user]# systemctl status syslog-ng
● syslog-ng.service - System Logger Daemon
   Loaded: loaded (/usr/lib/systemd/system/syslog-ng.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-04-07 11:56:50 UTC; 6min ago
     Docs: man:syslog-ng(8)
    Main PID: 674 (syslog-ng)
      Tasks: 3 (limit: 22798)
     Memory: 6.4M
        CPU: 63ms
    CGroup: /system.slice/syslog-ng.service
            └─674 /usr/sbin/syslog-ng -F -p /var/run/syslogd.pid

Apr 07 11:56:50 ip-172-31-0-46.ap-southeast-2.compute.internal systemd[1]: Starting System Logger Daemon...
Apr 07 11:56:50 ip-172-31-0-46.ap-southeast-2.compute.internal syslog-ng[674]: [2026-04-07T11:56:50.698735] WARNING: With use-dns(no), dns-ca-
Apr 07 11:56:50 ip-172-31-0-46.ap-southeast-2.compute.internal syslog-ng[674]: [2026-04-07T11:56:50.700750] Current position not matches to t
Apr 07 11:56:50 ip-172-31-0-46.ap-southeast-2.compute.internal systemd[1]: Started System Logger Daemon.
lines 1-15/15 (END)
```

- 3) Go to the directory below:

```
cd /etc/syslog-ng
```

- 4) List the files to check if syslog-ng.conf is available

```
ls
```

```
or
```

```
ll
```

```
[root@ip-172-31-0-46 ec2-user]# cd /etc/syslog-ng
[root@ip-172-31-0-46 syslog-ng]# ll
total 8
drwxr-xr-x. 2 root root  6 May  9  2025 conf.d
drwxr-xr-x. 2 root root  6 May  9  2025 patterndb.d
-rw-r--r--. 1 root root 1354 May  9  2025 scl.conf
-rw-r--r--. 1 root root 2345 May  9  2025 syslog-ng.conf
[root@ip-172-31-0-46 syslog-ng]#
```

- 5) Edit the syslog-ng.conf file

```
vi syslog-ng.conf
```

- 6) Add the below-listed lines to forward the data
Use your Splunk instance IP address & Port number:

```
source tcp_log_source {
file("/var/log/secure");
};

destination splunk_tcp_destination {
network("YOUR_SPLUNKHF_PRIVATEIP_ADDRESS" transport("tcp")
port(1050));
};

log {
source(tcp_log_source);
destination(splunk_tcp_destination);
};
```

```
@version: 3.35
@include "scl.conf"

# syslog-ng configuration file.
#
# This should behave pretty much like the original syslog on RedHat. But
# it could be configured a lot smarter.
#
# See syslog-ng(8) and syslog-ng.conf(5) for more information.
#
# Note: it also sources additional configuration files (*.conf)
#       located in /etc/syslog-ng/conf.d/

options {
flush_lines (0);
time_reopen (10);
log_fifo_size (1000);
chain_hostnames (off);
use_dns (no);
use_fqdn (no);
create_dirs (no);
keep_hostname (yes);
};

source s_sys {
system();
internal();
# udp(ip(0.0.0.0) port(514));
};

destination d_cons { file("/dev/console"); };
destination d_mesg { file("/var/log/messages"); };
destination d_auth { file("/var/log/secure"); };
destination d_mail { file("/var/log/maillog" flush_lines(10)); };
destination d_spool { file("/var/log/spooler"); };
destination d_boot { file("/var/log/boot.log"); };
destination d_cron { file("/var/log/cron"); };
destination d_kern { file("/var/log/kern"); };
destination d_mal { usertty("*"); };

filter f_kernel { facility(kern); };
filter f_default { level(info..emerg) and
not (facility(mail)
or facility(authpriv)
or facility(cron)); };
filter f_auth { facility(authpriv); };
filter f_mail { facility(mail); };
filter f_emergency { level(emerg); };
filter f_news { facility(uucp) or
(facility(news)
and level(crit..emerg)); };
filter f_boot { facility(local7); };
filter f_cron { facility(cron); };

#log { source(s_sys); filter(f_kernel); destination(d_cons); };
log { source(s_sys); filter(f_kernel); destination(d_kern); };
log { source(s_sys); filter(f_default); destination(d_mesg); };
log { source(s_sys); filter(f_auth); destination(d_auth); };
log { source(s_sys); filter(f_mail); destination(d_mail); };
log { source(s_sys); filter(f_emergency); destination(d_mal); };
log { source(s_sys); filter(f_news); destination(d_spool); };
log { source(s_sys); filter(f_boot); destination(d_boot); };
log { source(s_sys); filter(f_cron); destination(d_cron); };

# Source additional configuration files (.conf extension only)
@include "/etc/syslog-ng/conf.d/*.conf"

source tcp_log_source {
file("/var/log/secure");
};

destination splunk_tcp_destination {
network("172.31.0.83" transport("tcp") port(1050));
};

log {
source(tcp_log_source);
destination(splunk_tcp_destination);
};

# vim:ft=syslog-ng:ai:si:ts=4:sw=4:et:
~
~
:wq!
```

7) Check if 1050 port is enabled properly

```
semanage port -l | grep 1050
```

8) If the port is not enabled, Enable 1050 port in syslog using the below command:

```
semanage port -a -t http_port_t -p tcp 1050
```

```
[root@ip-172-31-0-46 syslog-ng]# semanage port -l | grep 1050
cma_port_t      tcp      1050
cma_port_t      udp      1050
[root@ip-172-31-0-46 syslog-ng]# semanage port -a -t http_port_t -p tcp 1050
Port tcp/1050 already defined, modifying instead
[root@ip-172-31-0-46 syslog-ng]# semanage port -l | grep 1050
cma_port_t      tcp      1050
cma_port_t      udp      1050
http_port_t     tcp      1050, 80, 81, 443, 488, 8008, 8009, 8443, 9000
[root@ip-172-31-0-46 syslog-ng]#
```

9) Restart syslog-ng service

```
systemctl restart syslog-ng
```

```
[root@ip-172-31-0-46 syslog-ng]# hostnamectl set-hostname LinuxSyslog
[root@ip-172-31-0-46 syslog-ng]# systemctl restart syslog-ng
```

10) Troubleshooting (if any):

```
egrep -i 'syslog-ng.*' /var/log/messages
```

In case if you see **Permission denied** in the logs, follow the following fixes:

i) **Temporary Fix:**

Run the following Commands:

```
getenforce
```

Expected Output:

```
Enforcing
```

```
setenforce Permissive
```

```
getenforce
```

Expected Output:

```
Permissive
```

ii) **Permanent Fix:**

Edit file:

```
vi /etc/selinux/config
```

Change:

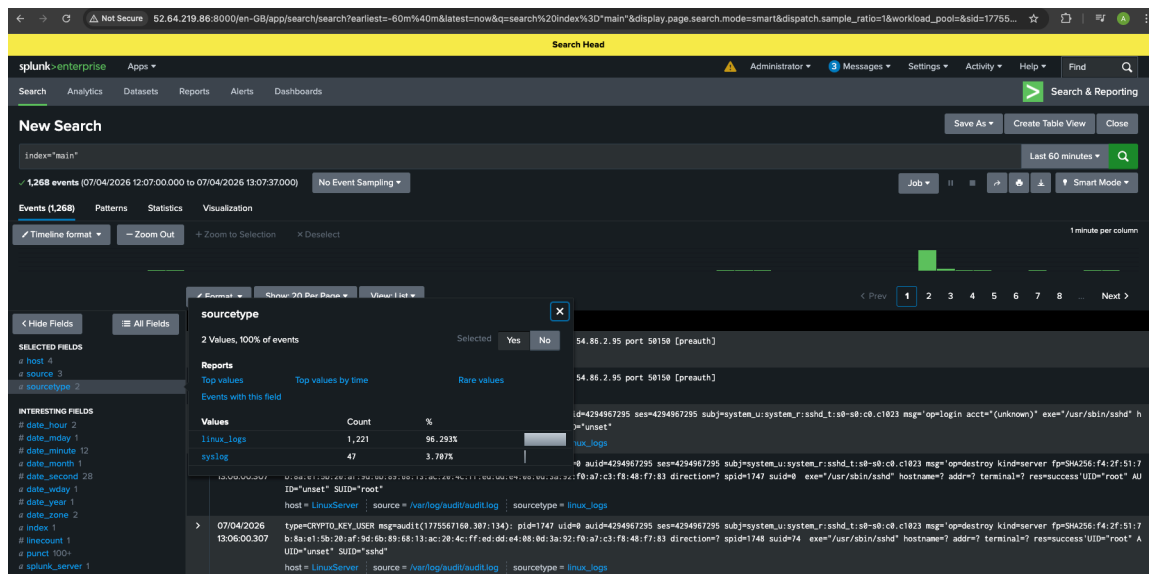
```
SELINUX=enforcing
```

to:

```
SELINUX=permissive
```


4. Verify the receiving events

On Search Head UI,
Go to Search & Reporting app,
index=main



Next Steps

Onboard Data from Windows Server – Refer
09_Onboard_data_from_Win_Server_n_Client_V1.pdf.